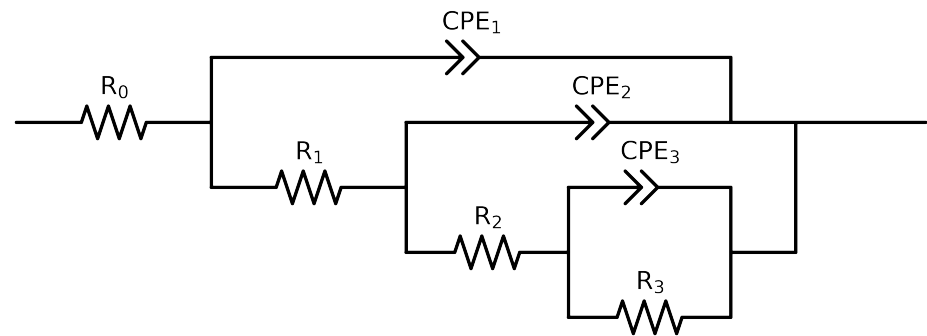


Comparative EIS Kinetics Report

Model Structure: R0-p(R1-p(R2-p(R3,CPE3),CPE2),CPE1)

Used data: original data



Element	Unit	Bounds	PNA_1_3_0h	PNA_1_3_1h
R ₀	Ω	[1.0e + 00, 1.0e + 03]	6.46e + 02 ± 2.1e + 00	4.82e + 02 ± 1.8e + 00
R ₁	Ω	[1.0e + 00, 1.0e + 08]	5.76e + 03 ± 6.3e + 02	6.41e + 03 ± 4.3e + 02
R ₂	Ω	[1.0e + 00, 1.0e + 08]	9.91e + 04 ± 1.5e + 04	7.87e + 04 ± 8.0e + 03
R ₃	Ω	[1.0e + 00, 1.0e + 20]	1.98e + 19 ± 4.0e - 01	3.98e + 08 ± 1.2e - 01
Q ₃	Ω ⁻¹ sec ^a	[1.0e - 09, 1.0e - 02]	1.06e - 06 ± 1.4e - 07	1.33e - 06 ± 1.0e - 07
n ₃	-	[5.0e - 01, 1.0e + 00]	8.56e - 01 ± 2.6e - 02	8.29e - 01 ± 1.5e - 02
Q ₂	Ω ⁻¹ sec ^a	[1.0e - 09, 1.0e - 02]	1.10e - 06 ± 1.7e - 07	1.04e - 06 ± 1.1e - 07
n ₂	-	[5.0e - 01, 1.0e + 00]	7.65e - 01 ± 2.3e - 02	7.76e - 01 ± 1.7e - 02
Q ₁	Ω ⁻¹ sec ^a	[1.0e - 09, 1.0e - 03]	9.06e - 07 ± 7.4e - 08	6.59e - 07 ± 3.9e - 08
n ₁	-	[5.0e - 01, 1.0e + 00]	8.20e - 01 ± 8.2e - 03	8.41e - 01 ± 6.0e - 03
χ ²	-	-	7.990e - 05	1.091e - 04

Analysis Results: PNA_1_3_0h

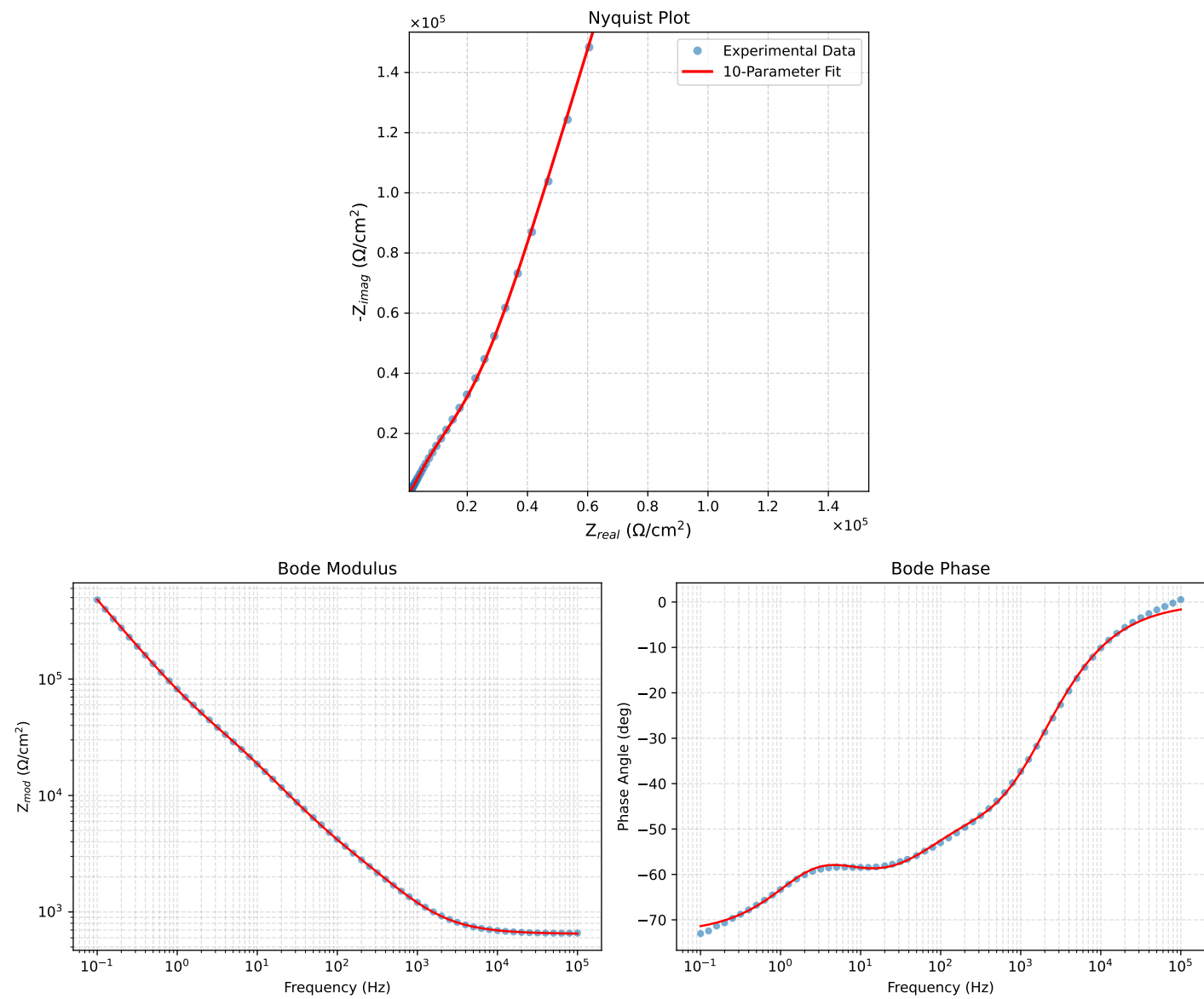


Figure 1: EIS Fit Results for PNA_1_3_0h

Fitted Data: PNA_1_3_0h

Frequency (Hz)	Z_{real} (Ω)	$-Z_{imag}$ (Ω)	Z_{mod} (Ω)	Z_{phase} ($^{\circ}$)
1.0002e+05	6.5171e+02	1.8570e+01	6.5198e+02	-1.6322
7.9512e+04	6.5286e+02	2.2407e+01	6.5324e+02	-1.9657
6.3105e+04	6.5425e+02	2.7070e+01	6.5481e+02	-2.3693
5.0215e+04	6.5593e+02	3.2631e+01	6.5674e+02	-2.8480
3.9902e+04	6.5798e+02	3.9373e+01	6.5915e+02	-3.4245
3.1699e+04	6.6047e+02	4.7512e+01	6.6218e+02	-4.1146
2.5137e+04	6.6354e+02	5.7407e+01	6.6601e+02	-4.9447
1.9980e+04	6.6724e+02	6.9212e+01	6.7082e+02	-5.9220
1.5879e+04	6.7177e+02	8.3433e+01	6.7693e+02	-7.0798
1.2598e+04	6.7737e+02	1.0068e+02	6.8481e+02	-8.4543
1.0020e+04	6.8417e+02	1.2119e+02	6.9482e+02	-10.0452
7.9282e+03	6.9274e+02	1.4641e+02	7.0804e+02	-11.9340
6.3079e+03	7.0308e+02	1.7598e+02	7.2477e+02	-14.0522
5.0347e+03	7.1566e+02	2.1080e+02	7.4606e+02	-16.4124
3.9931e+03	7.3170e+02	2.5353e+02	7.7438e+02	-19.1110
3.1441e+03	7.5237e+02	3.0626e+02	8.1232e+02	-22.1491
2.5195e+03	7.7624e+02	3.6427e+02	8.5746e+02	-25.1393
2.0008e+03	8.0710e+02	4.3549e+02	9.1710e+02	-28.3500
1.5807e+03	8.4655e+02	5.2137e+02	9.9422e+02	-31.6279
1.2536e+03	8.9494e+02	6.2043e+02	1.0890e+03	-34.7321
1.0007e+03	9.5306e+02	7.3221e+02	1.2019e+03	-37.5341
7.9003e+02	1.0280e+03	8.6753e+02	1.3451e+03	-40.1622
6.2881e+02	1.1159e+03	1.0173e+03	1.5100e+03	-42.3531
5.0223e+02	1.2194e+03	1.1848e+03	1.7002e+03	-44.1751
4.0015e+02	1.3423e+03	1.3761e+03	1.9224e+03	-45.7129
3.1550e+02	1.4916e+03	1.6031e+03	2.1897e+03	-47.0638
2.5202e+02	1.6527e+03	1.8471e+03	2.4786e+03	-48.1797
2.0032e+02	1.8378e+03	2.1322e+03	2.8149e+03	-49.2407
1.5801e+02	2.0522e+03	2.4739e+03	3.2143e+03	-50.3240
1.2556e+02	2.2846e+03	2.8621e+03	3.6621e+03	-51.4020
9.9734e+01	2.5465e+03	3.3210e+03	4.1850e+03	-52.5196
7.9449e+01	2.8404e+03	3.8585e+03	4.7912e+03	-53.6414
6.2919e+01	3.1882e+03	4.5147e+03	5.5269e+03	-54.7711
4.9867e+01	3.5945e+03	5.2943e+03	6.3993e+03	-55.8260
3.8422e+01	4.1417e+03	6.3445e+03	7.5767e+03	-56.8633
3.1250e+01	4.6622e+03	7.3290e+03	8.6862e+03	-57.5386
2.4934e+01	5.3402e+03	8.5802e+03	1.0106e+04	-58.1023
2.0032e+01	6.1291e+03	9.9872e+03	1.1718e+04	-58.4630
1.5625e+01	7.2123e+03	1.1840e+04	1.3864e+04	-58.6531
1.2467e+01	8.3979e+03	1.3782e+04	1.6139e+04	-58.6438
9.9734e+00	9.7799e+03	1.5961e+04	1.8719e+04	-58.5021
7.9719e+00	1.1391e+04	1.8431e+04	2.1667e+04	-58.2829
6.3516e+00	1.3256e+04	2.1261e+04	2.5055e+04	-58.0565
5.0134e+00	1.5433e+04	2.4613e+04	2.9052e+04	-57.9108
3.9860e+00	1.7750e+04	2.8350e+04	3.3448e+04	-57.9497
3.1758e+00	2.0224e+04	3.2664e+04	3.8418e+04	-58.2353
2.5161e+00	2.2939e+04	3.7908e+04	4.4308e+04	-58.8211
1.9955e+00	2.5840e+04	4.4204e+04	5.1203e+04	-59.6908
1.5879e+00	2.8951e+04	5.1752e+04	5.9300e+04	-60.7771
1.2581e+00	3.2468e+04	6.1158e+04	6.9242e+04	-62.0371
9.9777e-01	3.6443e+04	7.2635e+04	8.1265e+04	-63.3557
7.9274e-01	4.1011e+04	8.6559e+04	9.5783e+04	-64.6484
6.2903e-01	4.6416e+04	1.0364e+05	1.1356e+05	-65.8736
4.9888e-01	5.2863e+04	1.2447e+05	1.3523e+05	-66.9885
3.9860e-01	6.0329e+04	1.4890e+05	1.6065e+05	-67.9435
3.1672e-01	6.9511e+04	1.7913e+05	1.9214e+05	-68.7915
2.5148e-01	8.0626e+04	2.1582e+05	2.3039e+05	-69.5155
1.9998e-01	9.3959e+04	2.5984e+05	2.7630e+05	-70.1197
1.5879e-01	1.1018e+05	3.1330e+05	3.3211e+05	-70.6253
1.2601e-01	1.2985e+05	3.7801e+05	3.9969e+05	-71.0427
1.0016e-01	1.5346e+05	4.5550e+05	4.8065e+05	-71.3814

Analysis Results: PNA_1_3_1h

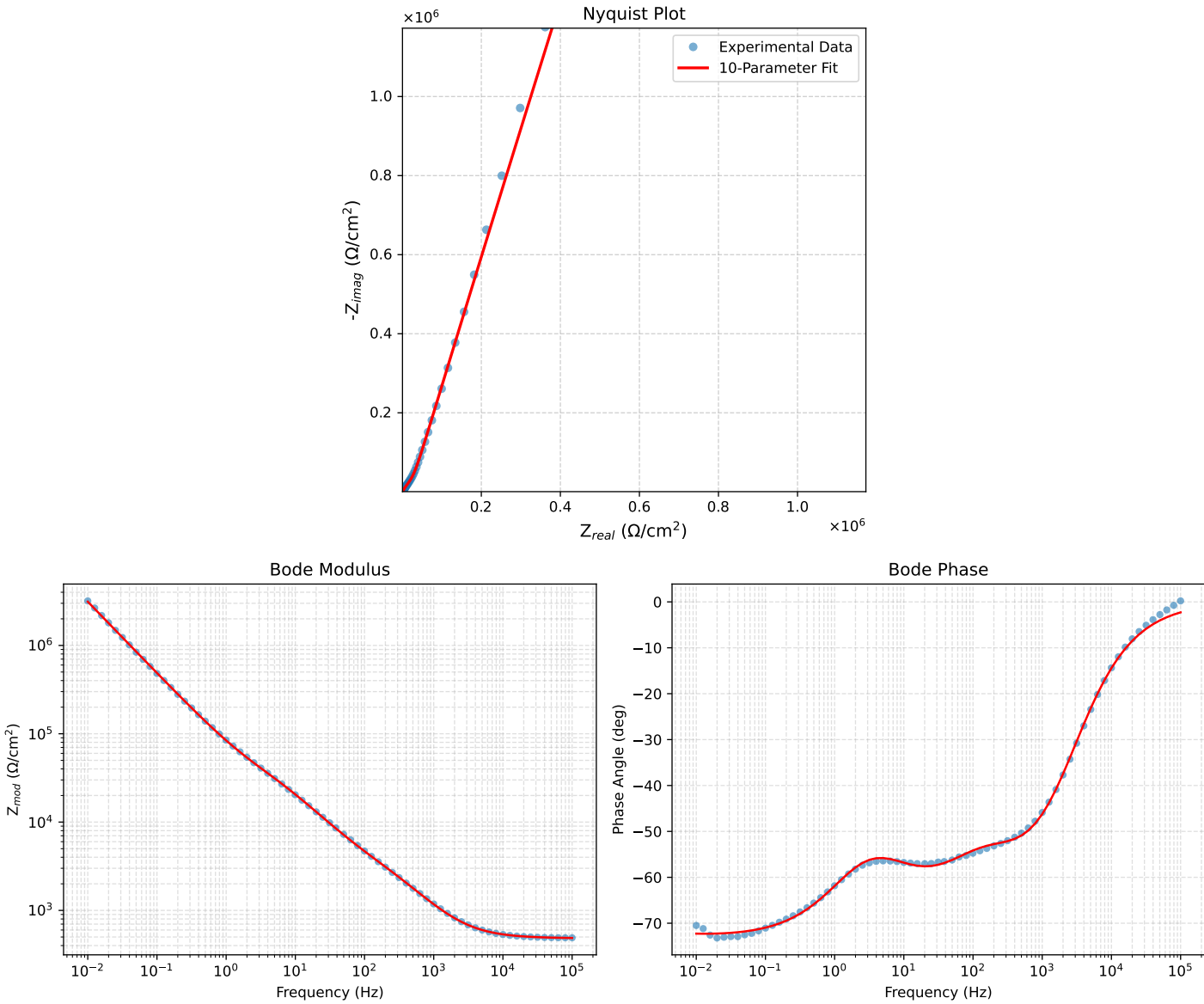


Figure 2: EIS Fit Results for PNA_1_3_1h

Fitted Data: PNA_1_3_1h

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0002e+05	4.8703e+02	1.9423e+01	4.8741e+02	-2.2838
7.9512e+04	4.8811e+02	2.3552e+01	4.8867e+02	-2.7624
6.3105e+04	4.8943e+02	2.8595e+01	4.9027e+02	-3.3437
5.0215e+04	4.9103e+02	3.4640e+01	4.9225e+02	-4.0352
3.9902e+04	4.9300e+02	4.2006e+01	4.9479e+02	-4.8701
3.1699e+04	4.9541e+02	5.0943e+01	4.9802e+02	-5.8710
2.5137e+04	4.9839e+02	6.1865e+01	5.0222e+02	-7.0758
1.9980e+04	5.0202e+02	7.4963e+01	5.0759e+02	-8.4927
1.5879e+04	5.0649e+02	9.0824e+01	5.1457e+02	-10.1663
1.2598e+04	5.1205e+02	1.1016e+02	5.2377e+02	-12.1417
1.0020e+04	5.1886e+02	1.3329e+02	5.3571e+02	-14.4070
7.9282e+03	5.2752e+02	1.6187e+02	5.5179e+02	-17.0590
6.3079e+03	5.3807e+02	1.9557e+02	5.7251e+02	-19.9747
5.0347e+03	5.5103e+02	2.3548e+02	5.9924e+02	-23.1387
3.9931e+03	5.6774e+02	2.8472e+02	6.3514e+02	-26.6336
3.1441e+03	5.8956e+02	3.4584e+02	6.8351e+02	-30.3960
2.5195e+03	6.1508e+02	4.1345e+02	7.4112e+02	-33.9084
2.0008e+03	6.4855e+02	4.9690e+02	8.1703e+02	-37.4583
1.5807e+03	6.9200e+02	5.9805e+02	9.1462e+02	-40.8344
1.2536e+03	7.4620e+02	7.1523e+02	1.0336e+03	-43.7863
1.0007e+03	8.1237e+02	8.4791e+02	1.1743e+03	-46.2262
7.9003e+02	8.9913e+02	1.0088e+03	1.3513e+03	-48.2898
6.2881e+02	1.0027e+03	1.1868e+03	1.5537e+03	-49.8047
5.0223e+02	1.1264e+03	1.3850e+03	1.7853e+03	-50.8800
4.0015e+02	1.2752e+03	1.6101e+03	2.0539e+03	-51.6204
3.1550e+02	1.4577e+03	1.8743e+03	2.3744e+03	-52.1267
2.5202e+02	1.6556e+03	2.1547e+03	2.7173e+03	-52.4627
2.0032e+02	1.8831e+03	2.4781e+03	3.1124e+03	-52.7684
1.5801e+02	2.1456e+03	2.8613e+03	3.5764e+03	-53.1344
1.2556e+02	2.4283e+03	3.2926e+03	4.0911e+03	-53.5911
9.9734e+01	2.7441e+03	3.7996e+03	4.6869e+03	-54.1627
7.9449e+01	3.0959e+03	4.3914e+03	5.3730e+03	-54.8163
6.2919e+01	3.5099e+03	5.1124e+03	6.2013e+03	-55.5283
4.9867e+01	3.9923e+03	5.9671e+03	7.1795e+03	-56.2151
3.8422e+01	4.6418e+03	7.1142e+03	8.4946e+03	-56.8769
3.1250e+01	5.2601e+03	8.1841e+03	9.7288e+03	-57.2701
2.4934e+01	6.0663e+03	9.5341e+03	1.1300e+04	-57.5325
2.0032e+01	7.0035e+03	1.11038e+04	1.3072e+04	-57.6047
1.5625e+01	8.2855e+03	1.2993e+04	1.5410e+04	-57.4741
1.2467e+01	9.6775e+03	1.5010e+04	1.7860e+04	-57.1891
9.9734e+00	1.1281e+04	1.7241e+04	2.0603e+04	-56.8035
7.9719e+00	1.3119e+04	1.9735e+04	2.3698e+04	-56.3845
6.3516e+00	1.5207e+04	2.2560e+04	2.7207e+04	-56.0167
5.0134e+00	1.7594e+04	2.5888e+04	3.1300e+04	-55.7993
3.9860e+00	2.0084e+04	2.9598e+04	3.5768e+04	-55.8408
3.1758e+00	2.2704e+04	3.3898e+04	4.0798e+04	-56.1867
2.5161e+00	2.5550e+04	3.9155e+04	4.6754e+04	-56.8742
1.9955e+00	2.8580e+04	4.5498e+04	5.3730e+04	-57.8650
1.5879e+00	3.1824e+04	5.3123e+04	6.1926e+04	-59.0760
1.2581e+00	3.5492e+04	6.2634e+04	7.1991e+04	-60.4615
9.9777e-01	3.9637e+04	7.4239e+04	8.4157e+04	-61.9020
7.9274e-01	4.4388e+04	8.8309e+04	9.8838e+04	-63.3140
6.2903e-01	4.9990e+04	1.0556e+05	1.1680e+05	-64.6583
4.9888e-01	5.6647e+04	1.2659e+05	1.3868e+05	-65.8919
3.9860e-01	6.4322e+04	1.5124e+05	1.6435e+05	-66.9599
3.1672e-01	7.3727e+04	1.8175e+05	1.9613e+05	-67.9200
2.5148e-01	8.5074e+04	2.1878e+05	2.3473e+05	-68.7508
1.9998e-01	9.8649e+04	2.6320e+05	2.8108e+05	-69.4539
1.5879e-01	1.1513e+05	3.1718e+05	3.3743e+05	-70.0507
1.2601e-01	1.3508e+05	3.8253e+05	4.0568e+05	-70.5502
1.0016e-01	1.5902e+05	4.6080e+05	4.8747e+05	-70.9607
7.9503e-02	1.8812e+05	5.5677e+05	5.8674e+05	-71.2995
6.3140e-02	2.2323e+05	6.7003e+05	7.0624e+05	-71.5737
5.0123e-02	2.6578e+05	8.0806e+05	8.5064e+05	-71.7934
3.9833e-02	3.1701e+05	9.7361e+05	1.0239e+06	-71.9649
3.1638e-02	3.7913e+05	1.1735e+06	1.2332e+06	-72.0957
2.5126e-02	4.5439e+05	1.4145e+06	1.4857e+06	-72.1906
1.9957e-02	5.4551e+05	1.7045e+06	1.7897e+06	-72.2531
1.5849e-02	6.5606e+05	2.0540e+06	2.1562e+06	-72.2862
1.2590e-02	7.9001e+05	2.4742e+06	2.5973e+06	-72.2917
1.0001e-02	9.5273e+05	2.9800e+06	3.1286e+06	-72.2703